

Membership Inference Attack Report

1 Objective

The objective of this experiment was to understand and evaluate the vulnerability of a trained machine learning model to Membership Inference Attacks (MIA). A Membership Inference Attack is a privacy attack where an adversary attempts to determine whether a specific data sample was part of the training dataset of a target model, based solely on its outputs (e.g., confidence scores).

The experiment was conducted on a SimpleCNN model trained on CIFAR-10, using the `membership_stable.py` script. This report explains the methodology, attack setup, evaluation metrics, and experimental results in detail.

2 Background and Concept

When a model is trained on data, it tends to memorize the training samples to some degree. As a result:

- The model often outputs higher confidence for data it has seen (training members).
- It shows lower confidence or higher uncertainty for unseen samples (non-members).

A Membership Inference Attack (MIA) exploits this difference in output behavior. If the attacker can access the model's confidence scores (softmax probabilities), they can statistically decide whether a sample was part of the training set.

3 Methodology

3.1 Model and Dataset

- **Model:** SimpleCNN
 - Three convolutional blocks (Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool)
 - Two fully connected layers with dropout
- **Dataset:** CIFAR-10 (50,000 training and 10,000 test images)
- **Training Checkpoint:** `best_cnn_cifar10.pth`

- **Batch Size:** 128
- **Attack Signal:** Max softmax confidence (primary indicator)

3.2 Attack Pipeline

The attack was performed using the function `run_membership_attack()` in the script. It follows these steps:

1. **Collect Model Outputs:** The model is evaluated on both member and non-member samples. For each sample, softmax confidence, prediction entropy, and per-sample loss are computed.
2. **Compute Threshold (Youden’s J Statistic):** The attack chooses a decision threshold to separate members and non-members:

$$J = TPR - FPR$$

where TPR is the true positive rate and FPR the false positive rate. The threshold maximizing J is selected.

3. **Evaluate the Attack:** Samples are classified as members if their confidence $\geq$ threshold. Metrics such as accuracy, precision, recall, F1, AUC, and J are then computed.

4 Evaluation Metrics Explained

Metric	Meaning	Formula / Interpretation
Accuracy	Overall correctness	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	True members among predicted members	$\frac{TP}{TP+FP}$
Recall (TPR)	True members correctly predicted	$\frac{TP}{TP+FN}$
F1-score	Harmonic mean of Precision and Recall	$\frac{2PR}{P+R}$
AUC	Separability (1.0 perfect, 0.5 random)	Computed via ROC curve
Youden’s J	Optimal threshold separation	$J = TPR - FPR$

5 Experimental Results

The attack was executed with:

```
python3 membership_stable.py
```

Metric	Value
Threshold (Confidence)	0.8212
Youden’s J	0.0618
Accuracy	0.6970 (69.7%)
Precision	0.8300 (83.0%)
Recall	0.7919 (79.2%)
F1-score	0.8105 (81.1%)
AUC	0.5148

6 Result Interpretation

6.1 Model Behavior

The model outputs higher confidence for training samples than for test samples. The threshold of 0.8212 best separates the two according to Youden’s J.

6.2 Attack Effectiveness

The attack achieves 69.7% accuracy, meaning it correctly identifies membership roughly 7 out of 10 times—better than random guessing (50%), but far from perfect. High precision (0.83) and recall (0.79) indicate the attack is somewhat reliable, though the AUC near 0.51 shows weak separability.

6.3 Privacy Interpretation

These results indicate partial but weak membership leakage. The model leaks some information about whether a data point was in the training set, but not enough to enable a strong attack.

6.4 Trade-off Between Attack Success Rate and Metric

There exists a clear trade-off between the *attack success rate* and the chosen evaluation metric. When the attacker adjusts the decision threshold to maximize one metric (e.g., precision or recall), others are affected:

- A higher threshold increases precision but lowers recall—fewer false positives but more missed members.
- A lower threshold increases recall but decreases precision—more members detected but more false alarms.
- The Youden’s J statistic offers a balanced compromise by maximizing the difference between true and false positive rates, but may not maximize absolute accuracy.

- The AUC reflects overall separability independent of threshold; in this experiment, its near-random value (0.51) suggests that threshold tuning cannot substantially improve success rate.

Thus, improving attack success requires either more informative confidence features or a more discriminative attack model, as mere threshold adjustment yields diminishing returns due to the overlapping score distributions.

7 Conclusion

The Membership Inference Attack using max softmax confidence achieved:

- Accuracy: 69.7%
- Precision: 83%
- Recall: 79%
- F1-score: 81%
- AUC: 0.51

These results show that while some privacy leakage exists, it remains limited. The trade-off analysis further confirms that attack performance is bounded by the weak separability of member and non-member confidence distributions, maintaining reasonable robustness against MIA threats.